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RESEARCH ARTICLE

ABDviaMSIFAT: Abnormal Crowd Behavior Detection Utilizing a Multi-Source Information Fusion Technique

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ABSTRACT Detecting abnormal crowd behavior in surveillance videos is a significant challenge due to the intricate and constantly evolving crowd dynamics. To solve this issue, we suggest a new method that combines data from various sources with different characteristics to enhance the precision of detecting human behavior in crowds. Our approach involves two separate pipelines that work simultaneously to produce scores for frames in a video segment. These scores are later modified for the individual level of the group, allowing for behavior recognition through the assessment of fuzzy logic functions. In the first pipeline, we utilize a depth-wise Separable Convolutional Neural Network (DWS-CNN) that provides reduced filtering compared to standard CNNs. The second pipeline combines the LiteFlowNet detector with the MOSSE tracker and a DSC-GRU network to generate high-level captions for objects in video frames. We implement the weighted average (WA) method to improve anomaly detection accuracy. Methods like weighted averages can mitigate the influence of outliers and noise in the outcomes or evaluations. Utilizing linguistic variables to represent scores and computing weighted averages of scores from two pipelines enhances the quality and reliability of these variables, creating fuzzy predicates that characterize people's movements, presence, and responses at a microscopic scale. Our approach exceeds conventional visual and motion-centric methods, enabling a more comprehensive grasp of abnormal behaviors. Our suggested approach outperforms state-of-the-art methods in terms of effectiveness and performance based on tests done on well-known datasets.

INDEX TERMS Crowd behavior recognition, information-sharing, fuzzy logic functions, averaging techniques.

I. INTRODUCTION

Real-time monitoring and analyzing captured frames in video surveillance systems is a significant area of research in computer vision. Despite the challenges, it remains an important open issue for researchers in this field. Understanding the activities that occur in the monitored area is crucial, and intelligent technology is needed to assist human operators in determining the conceptual nature of the detected activities. Recently, several mechanisms have been proposed for monitoring and analyzing activities using machine learning algorithms. These algorithms are utilized to identify and track a target, such as a person, car, or moving object, and classify their activities as normal or abnormal. The analysis of sudden collective events, such as mass escape, theft, or attacks, presents a new trend with numerous complex challenges. While many research studies have focused on the analysis, comprehension, or labeling of individual activities in monitoring videos, the conceptual analysis of group events remains an open topic [1]. The traditional method for determining whether a detected event in a video is an abnormal or normal action is to train learned models and classify any event that deviates from these anomalous models [2], [3]. However, this method is costly and requires enhancements to analyze all frames, resulting in high computational complexity, especially in the group activity diagnosis. It also necessitates

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focusing on processing and accounting for foreground pixels using ground separation algorithms. Moreover, these methods reduce the amount of the data involved in the diagnostic process, leading to false alarms in unclear images, and the diagnostic process for abnormal events requires processing all frames for a correct evaluation.

Flexible tools with various advantages allow for the modification and control of feature maps extracted from specifications. The research aims to enable machines to autonomously detect, understand, and explain activity in scenes without human intervention. Some studies use neural networks to extract high-level concepts from video events to identify anomalous actions [4]. Using high-level concepts from images can enhance conceptual understanding and improve classification processes [5].

Analyzing anomaly crowd scenes is difficult due to numerous people's movements in different directions, making it hard to identify useful features [6], [7]. A paper introduces a parallel framework using two pipelines to resolve this problem. The first pipeline uses DWS-CNN for lighter filtering [8], [9], while the second pipeline includes a reinforcement module with LiteFlowNet and LSTM to extract relevant captions. These captions are analyzed through an embedding network to create special features for the event description.

In this study, our main contributions are:

- The study identifies abnormal events in scenes by proposing components to provide critical information for determining their causes. It combines conceptual analysis with a lightweight deep network to detect collective abnormal events, reducing decision-making errors. The method incorporates DWS-CNN and LiteFlowNet descriptor features for improved anomaly identification in crowded scenes.
- A feedback loop enhances detection by adjusting weights between parallel pipelines to improve system performance. Membership functions and linguistic variables allow flexibility and clear analysis of abnormal events. Unlike traditional methods, input from multiple sources is considered for more accurate judgments. Overall, the study introduces a novel pipeline for detecting abnormal events at a higher level, improving the system's performance in identifying anomalies effectively.

The paper is organized as follows: Section II offers relevant analysis and background information on abnormal event detection. Section III introduces the proposed method. Section IV presents the study results based on empirical simulation and evaluates the proposed method. Finally, Section V provides the study summary, conclusion, and possible recommendations for future research.

II. RELATED WORKS

Previous research on abnormal event detection has primarily focused on identifying deviations from normal behavior, often overlooking the events' underlying causes or sources. While these methods can be effective for analyzing individual instances, they may not be suitable for complex scenarios involving crowded scenes and multiple events. It is crucial to determine whether an abnormal event in a crowded scene is genuinely anomalous or a result of interactions within the scene. Further research is needed to explore this aspect and develop methods that consider the underlying causes of abnormal events in crowded scenes.

In the realm of addressing challenges in this particular field, several notable works have emerged. These works can be classified into three categories: those that relied on traditional methods, those that integrated deep learning frameworks to achieve a comprehensive model, and the studies that harmoniously combined traditional algorithms with deep learning techniques.

A. TRADITIONAL MACHINE LEARNING TECHNICS

In [10], a spatiotemporal texture analysis approach is proposed, which converts two-dimensional space to a three-dimensional model for texture analysis. However, this method is influenced by crowd density in the scene, limiting its effectiveness. Santoro et al. [11] utilize a pyramidal Lucas-Kanade tracker and density-based clustering to analyze crowd behavior. While this algorithm shows promise, it is tested on a specific database and may not generalize well to other scenarios. Wang and Snoussi [12] introduce histograms of optical flow orientation (HOFO) for detecting abnormal behavior. This algorithm employs a one-class SVM frame classification, but its effectiveness may be limited to specific types of abnormal behavior.

B. DEEP LEARNING TECHNICS

Xu et al. in [13] and Ribeiro et al. [14] present methods using autoencoders for shape and motion learning and reconstructing error-based anomaly detection. However, these methods do not specifically focus on crowded scenarios and may have limitations in accurately detecting anomalies in such complex environments. An artificial intelligence-based technique for predicting and detecting robbery behavior potential in indoor surveillance is presented in [15]. Additionally, [16] proposes a system utilizing semantic segmentation for human motion analysis, and attraction force models for pedestrian counting and tracking, and a genetic algorithm for anomaly detection. While these methods show promise, their accuracy decreases significantly in large crowds due to overlap constraints, and they are sensitive to noise and the initial selection of cluster centers. In [17], the authors introduce a systematic study using the multi-scale motion consistency network (MSMC-Net) to address video anomaly detection (VAD) for crowd-level abnormal behaviors (CABs). However, CABs do not exhibit significant differences from normal behaviors when observed locally, which can make their detection more challenging. The scale of CABs can vary from one scenario to another, adding to the complexity of detecting them.

Rawanbakhsh et al. [18] propose an anomaly detection method that combines representative convolutional neural networks with low-level optical flow. Reference [19] presents a hybrid deep learning-based approach utilizing both deep and hand-crafted features. These methods show limitations in handling illumination changes and intricate crowded scenes, particularly in queue conditions. Finally, the related work discusses techniques for privacy-preserving anomaly detection [20], [21]. These approaches can benefit from incorporating fuzzy logic techniques to achieve a better balance between privacy and accuracy, and improved robustness in handling diverse scenarios.

Few papers or research reports address a knowledge-based approach to crowd modeling and categorization. In some recent work [17], [18], [22], Multi-agent infantry models were created for significant outdoor events [17], treating pedestrians as knowledgeable agents. Three models were developed based on crowd arrival, dispersal, and evacuation phases. In [25] a paradigm for crowd behavior recognition is proposed, comprising a multi-person tracker and a knowledge-based classifier using fuzzy predicates. In a similar way, abnormal crowd recognition method proposed in [26] analyzes small trajectories and macroscopic characteristics, including dense optical flow and histogram of optical flow orientation, at a microscopic level. It employs a fuzzy logic predicate-based approach to classify movement patterns by evaluating fuzzy predicates of individuals or groups. Reference [27] presents a fuzzy inference approach to crowd emotion assessment based on the arousal-valence model of crowd movement. The fuzzy inference system predicts arousal and valence values by extracting descriptors of crowd emotion such as enthalpy, magnitude variance, confusion index, and crowd density [28]. These methods used the fuzzy logic system, without semantic analysis and strengthened visual features by analyzing textual information. They did not exploit linguistic variables, which are the original tool in the fuzzy system, and it expresses abnormal cases well.

III. OUR WORK POSITIONING

To the best of our knowledge, there is a severe lack of literature that explores the integration of semantic analysis of captions into current methodologies. Our method goes beyond traditional visual and motion-based approaches, allowing for a deeper understanding of abnormal activities. This semantic context enhances the accuracy and reliability of anomaly detection in surveillance videos. We claim that the iterative feedback loop in our method plays a crucial role in refining the results and achieving better performance in object detection, tracking, and anomaly detection. This iterative process allows for continuous improvement by leveraging the outputs of each pipeline to inform and enhance subsequent iterations. Besides, adding weights in each iteration strengthens the anomaly detection process by assigning importance to different features and aspects of the surveillance videos. These weights enable the model to focus on relevant information and prioritize anomaly detection, leading to more accurate and reliable results. The importance of the weights lies in their ability to direct the model's attention towards the most informative and discriminative features, effectively reducing false positives and false negatives in anomaly detection. To further enhance the reduction of false alarms (false positives and false negatives), we employ the weighted average technique to adjust the weights assigned to the scores predicted by the two pipelines, helping to minimize the impact of outliers and noise in the results. This fine-tuning process ensures that the model learns to distinguish between normal and abnormal behaviors with higher precision and recall. Moreover, by combining the output of the pipelines with the fuzzy system, a more intelligent and adaptive system can be created that can handle uncertainty and make more informed decisions based on the detected abnormal behaviors.

IV. METHOD

Crowd behavior for identifying anomalous systems consists of two main pipelines running in parallel, see Fig. 1.

1. A neural network based on depth-wise separable convolutions (DS-CNN).
2. A 3-stage pipeline for multi-person tracking and activity analysis in the scene.

The initial phase of our framework involves a pre-processing module, which includes a series of steps to improve the quality of input data by standardizing, resizing, reducing noise, and extracting features. It aims to standardize data for analysis. Different techniques and algorithms may be used depending on system requirements, but the same goal remains preparing data for future system processes. This framework offers the following services: (i) Service 1: Parallel processing and information exchange. The first pipeline can focus on detecting anomaly events using a fuzzy system, while the second pipeline can focus on object detection, tracking, and caption generation. (ii) Service 2: Feedback loop integration. (iii) Service 3: Fuzzy system.

Fig. 2 shows all of these services and how they communicate with each other. Indeed, Service 1 is based on detecting and classifying basic abnormal behaviors such as sudden crowd gathering/dispersal. The output of this pipeline is a set of predictions for each frame. By using fuzzy logic, anomaly scores are generated for each video frame. Subsequently, the second pipeline would be employed for analysis of the contextual information and the exchanges among individuals or groups in the crowd, generating captions that depict the objects and activities. Service 2 is required to establish a feedback loop between the two pipelines to improve overall performance. The anomaly event scores from the first pipeline can serve as feedback to refine object detection and tracking in the second pipeline. Similarly, predicted captions from the second pipeline can provide feedback to enhance anomaly detection in the first pipeline. This iterative feedback loop helps refine results and achieve better performance.

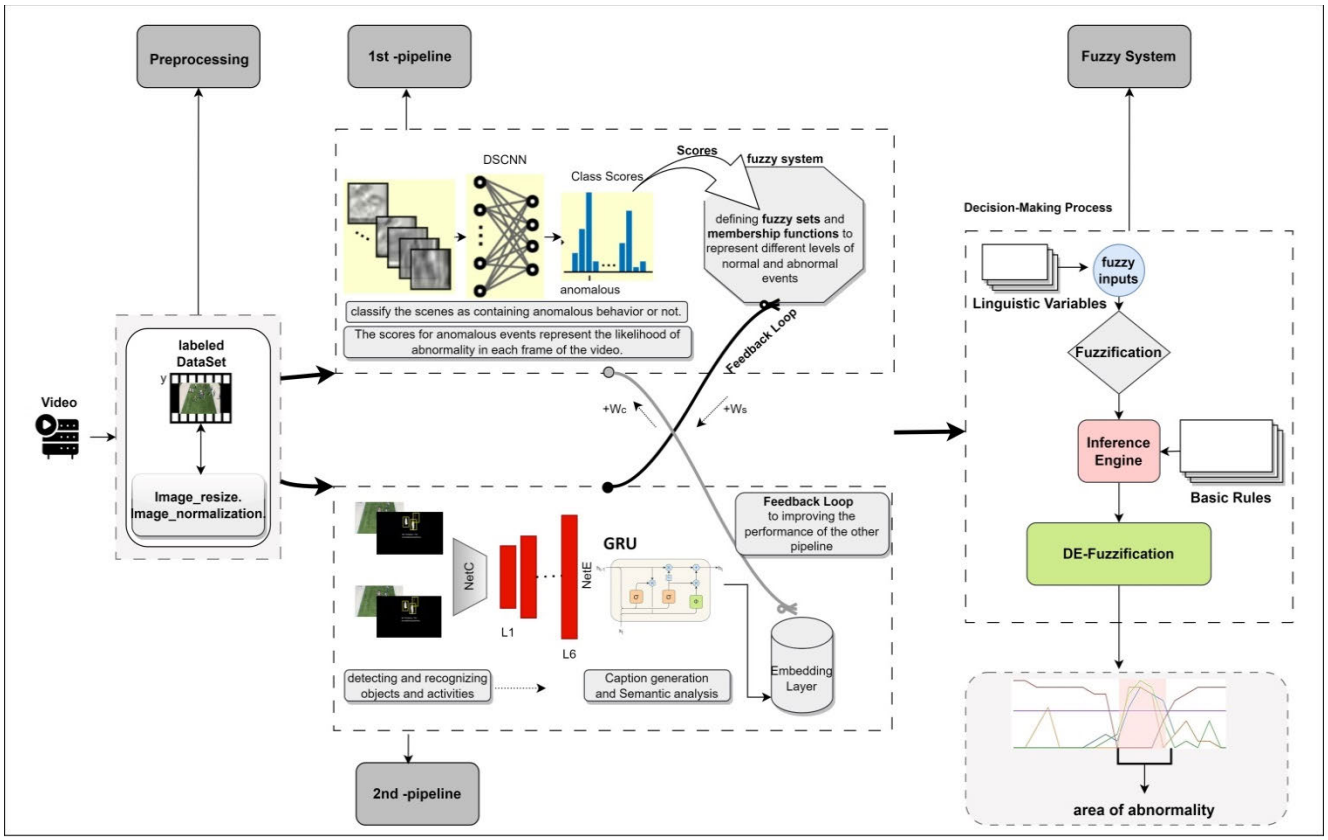


FIGURE 1. Infrastructure of the abnormal crowd behavior recognition system.

Integrating outputs of the two pipelines through parallel processing, fusion, or establishing a feedback loop strengthens both pipelines to improve overall results. Integration allows for a more comprehensive analysis of video frames, incorporating both anomaly event detection and high-level object-level information. For example, if the first pipeline detects a high score for an anomaly event in a particular frame, this information can be used to focus the object detection and tracking on the objects that are potentially related to the anomaly event. By prioritizing the detection and object tracking, the second pipeline can provide more accurate and relevant information about the anomaly event, improving the overall performance of the system. Service 3 is required to build the necessary procedures to determine whether a specific video clip has an abnormality. The procedures include:

Generating Linguistic Variables: Consider generating linguistic variables for the anomaly scores and the captions. For instance, the linguistic variable for anomaly scores can be defined as ''Low'', ''Medium'' and ''High'' while the linguistic variable for captions can be defined as ''Normal'' and ''Abnormal''.

Creating a Set of Fuzzy Rules: It establishes a relationship between the anomaly scores and the captions. These rules are based on the correlation between the anomaly scores and abnormal activities described in the captions. For example, IF the anomaly score is ''High'' and the caption is ''Abnormal'' then the video clip contains an anomaly. IF the anomaly score is ''Low'' or caption is ''Normal'' then the video clip does not contain an anomaly.

Applying Fuzzy Inference: The defined linguistic variables and fuzzy rules are utilized to determine the membership level of the video clip containing an anomaly consisting of fuzzification, rule evaluation, and defuzzification.

Performing Defuzzification: Defuzzification is performed to obtain a clear value representing the final decision on whether the video clip contains an anomaly. This can be achieved using methods such as maximum membership.

Establishing a Threshold Value: A threshold value determines the minimum degree of membership needed for the final decision. If the clear value obtained from defuzzification surpasses the threshold, it signifies the presence of an anomaly in the video clip.

A. DSCNN-BASED MODEL LEARNING (1ST PIPELINE)

We divided the video fed into our network into two epochs: Epoch_n for normal and Epoch_{an} for anomaly segments. Each epoch represents a collection of instances, where a positive video is represented by a positive epoch consisting of different temporal segments $(S_1, S_2, S_3, \ldots, S_k)$. A negative video is denoted by a negative epoch consisting of negative instances

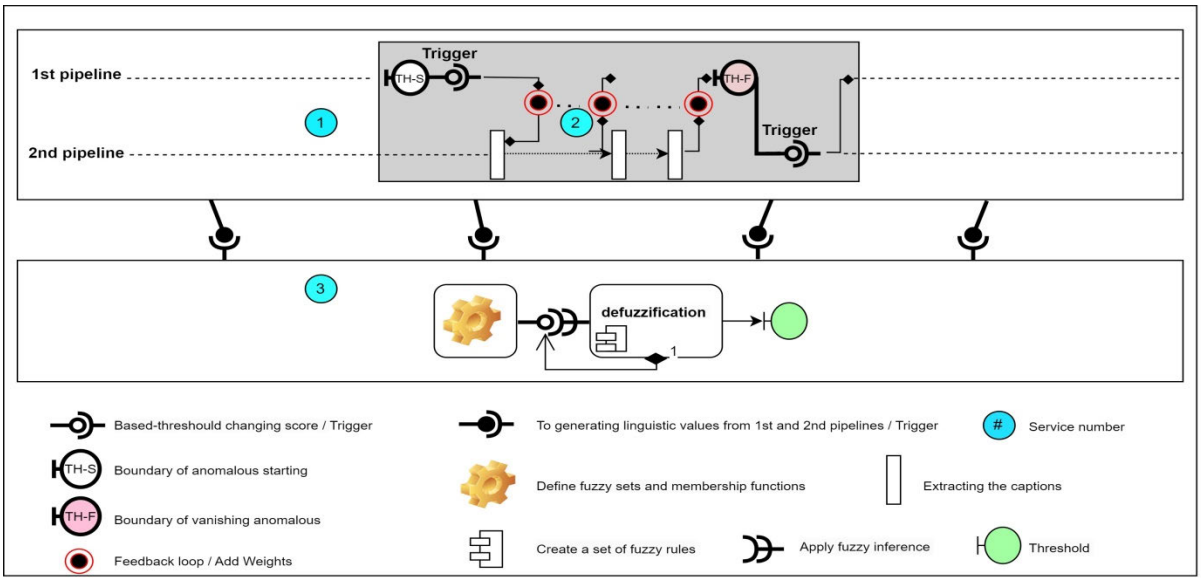


FIGURE 2. The component diagram of the logical architecture view of the proposed framework.

(N_1, N_2, N_3, …, N_k) containing the anomaly. Multi-instance learning is obtains temporal annotations for epochs by using video-level labels [29].

The architecture of DSCNN with 16 layers is depicted in Fig. 3.

The specifications for each layer are 16 (3, 3) kernels in the first layer, 32 in the second, 64 (4, 4) in the third, and 128 in the fourth.

Each layer extracts unique features from the input image, enabling the model to learn hierarchical representations [30].

Max pooling reduces feature map dimensions, dropout prevents overfitting, and batch normalization speeds training. Upon conducting experiments on this structure, we found that implementing Dropout following the Convolution layer helps the model find improved solutions and learn robust features [31]. Including Batch Normalization following Dropout enhances the training process by normalizing the inputs of the layers, leading to improved speed and stability [32]. We found a suitable configuration for effective feature extraction, regularization, and normalization to manage the model's complexity and improve its generalization.

We have a classical binary classification cross-entropy loss:

$$L\left(y, y^{\wedge}\right) = -\sum_{i=1}^{n}\left(y_i.logy_i^{\wedge} + (1-y_i).\log[1-y_i^{\wedge}]\right) \tag{1}$$

where in (1), y represents the ground truth, $y^{\wedge}$ is prediction and $L\left(y, y^{\wedge}\right)$ is the loss function.

The objective function can be optimized in relation to the highest-scored instance in that epoch:

$$L\left(y, y^{\wedge}\right) = -\sum_{i=1}^{n}\left(Y_{EPOCH_k}.logy_i^{\wedge} + \left(1-Y_{EPOCH_k}\right).\log[1-y_i^{\wedge}]\right) \tag{2}$$

where Y_{EPOCH_k} represents the epoch-level label, k represents the total number of epochs, and all other variables are the same as in (1).

We build additional organization or value in the loss for the best score. These types of additions further reinforce the learning process in the proposed network.

$$\gamma \sum_j |W_j|\gamma \geq 0 \tag{3}$$

We set $\gamma = 0.01$, to create a new layer with this addition and insert the new layer before the last layer.

Finally, the objective function is given by:

$$L_{final}(W) = L\left(y, y^{\wedge}\right) + \gamma \sum_j |W_j|\gamma \geq 0 \tag{4}$$

B. RECOGNIZING HUMAN ACTIVITIES (2ND PIPELINE)

The same segment instances fed into the first pipeline are fed into the second pipeline to improve the result of the first pipeline and add further parameters for developing our knowledge set. By modeling the background, isolating moving objects, tracking their mobility and calculating their activity, we hope to find the people participating in the activities depicted in the movie.

1) BACKGROUND SUBTRACTION

At this stage, we follow [22], [33], [34], and [44] with some additions that apply to our case. First, an adaptive Gaussian mixture model (AGMM) is used for background subtraction. The AGMM has a low computational cost and is highly resistant to illumination changes, resulting in higher accuracy. In addition, the number of Gaussian mixture components (GMC) in the AGMM can be adaptively adjusted depending on the frequency of pixel change. Assuming that ($Px_{i,t}$) is the current value of pixel i at time t and ($PPx_{i,t-1}$) is the previous

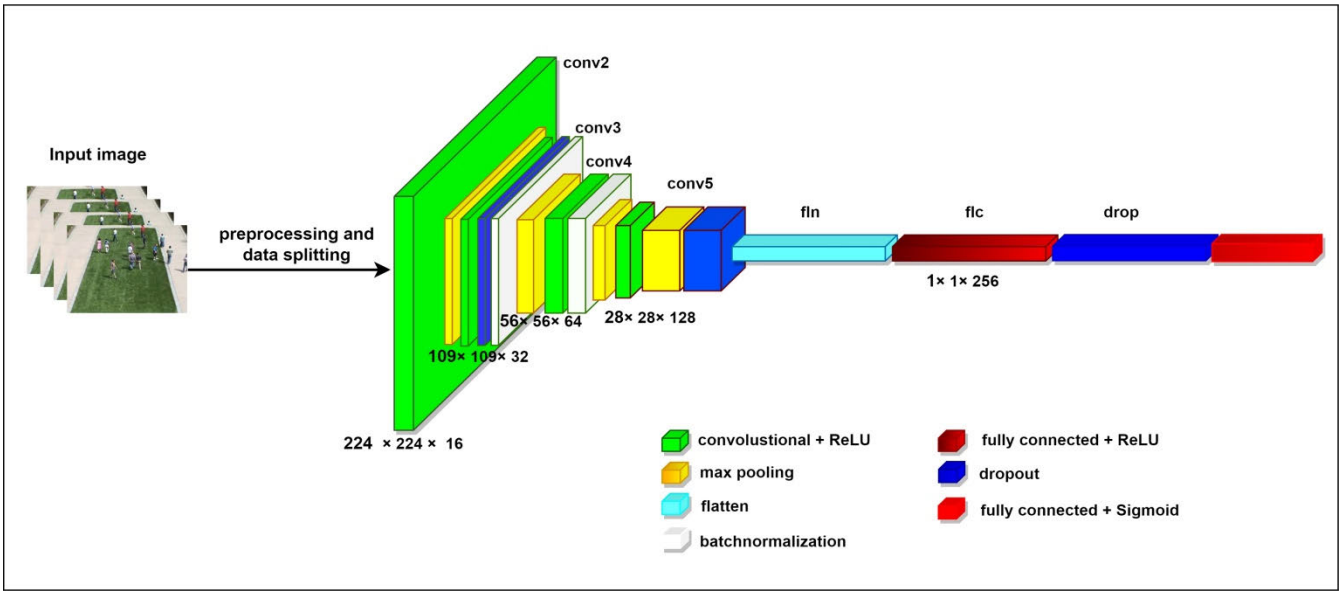


FIGURE 3. The DSCNN-based architecture uses convolutional and pooling layers for feature extraction and dimension reduction. Dropout and batch normalization layers are included to prevent overfitting and enhance training efficiency. The final classification is done by fully connected layers.

average value for the same. Equation (5) is used to determine changed pixels (foreground). According to the formula, the pixel i is a foreground pixel. Otherwise, it is considered as background.

$$\left|\mathrm{PX}_{i,t} - \mathrm{PPx}_{i,t-1}\right| > \mathrm{TRESHOLD} \tag{5}$$

We can generalize this in cases where the background should be verified regularly (every time an image changes) so that newly appearing objects are considered part of the updated or verified background. In other words, the background is updated for detecting human-like patches using (6).

$$B_i = (1-r) \times I_i + r \times B_{i-1} \tag{6}$$

This update ensures that new objects are accurately identified as part of the background.

In this method, B_{i-1} stands for the previous background, B_i for the current updated background, I_i for the current image information, and r defines the updating rate. The region-of-interest algorithm generates a binary (black or white) image in which the extracted patches are 1s (i.e., white) and the background is 0s (i.e., black). Since edges are easy to recognize in binary images, the ratio of width to height of each patch can be easily calculated. The algorithm excludes non-human patches whose width-to-height ratio exceeds a certain threshold (TH = 0.5 in this algorithm). Non-human objects with a width-to-height ratio greater than 0.5, such as cars, bicycles, animals, or buses, are excluded. Fig. 4 shows how this algorithm works with a UMN dataset sequence.

Yes, sometimes we may miss identifying cases that are not human, but we have utilized this method due to its speed since it provided rapid outcomes in identifying human cases [33]. Someone can be asleep or in a state of prostration, in which case the ratio will be over 0.5. However, this situation is

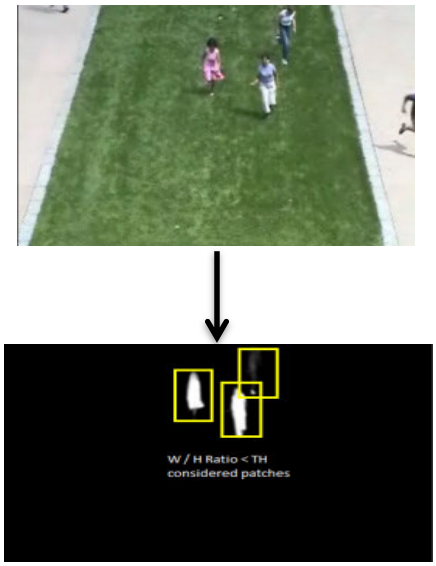


FIGURE 4. AGM patches – considering the UMN dataset.

beyond our scope, and we are not interested in evaluating its efficacy in this study. We discovered from our testing on different objects, that a person can be identified at a ratio of 0.5 or lower based on our experiences. For example, when observing a group containing a person in a wheelchair, his stem will move, and the width-to-length ratio is less than 0.5 or close to 0.5. In contrast, for animals, cars, or buses, the ratio exceeds 0.5.

Following the methodology used in [33] and [34], we continue our work after obtaining the (ROI) to identify human activities. For this purpose, we use an elementary lightweight

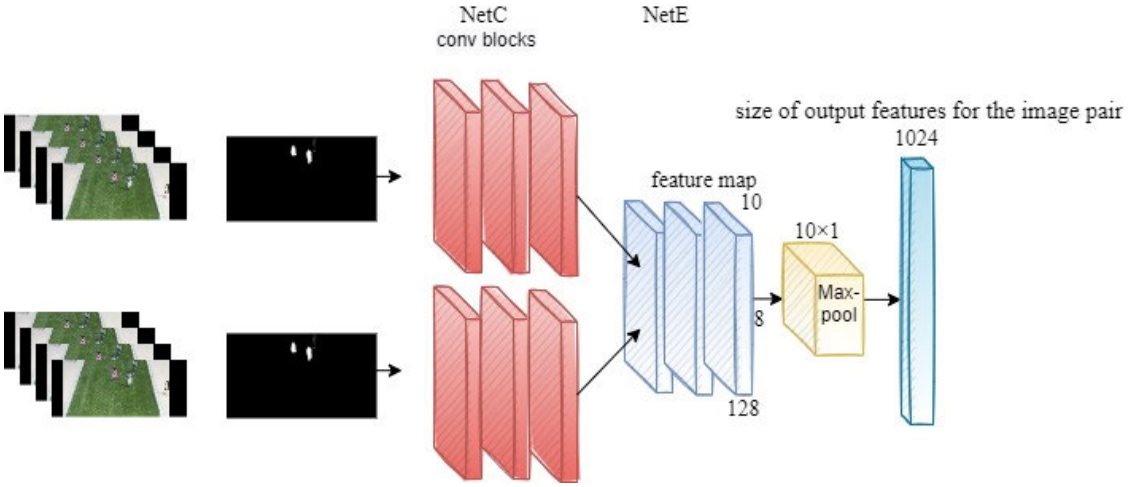


FIGURE 5. LiteFlowNet architecture.

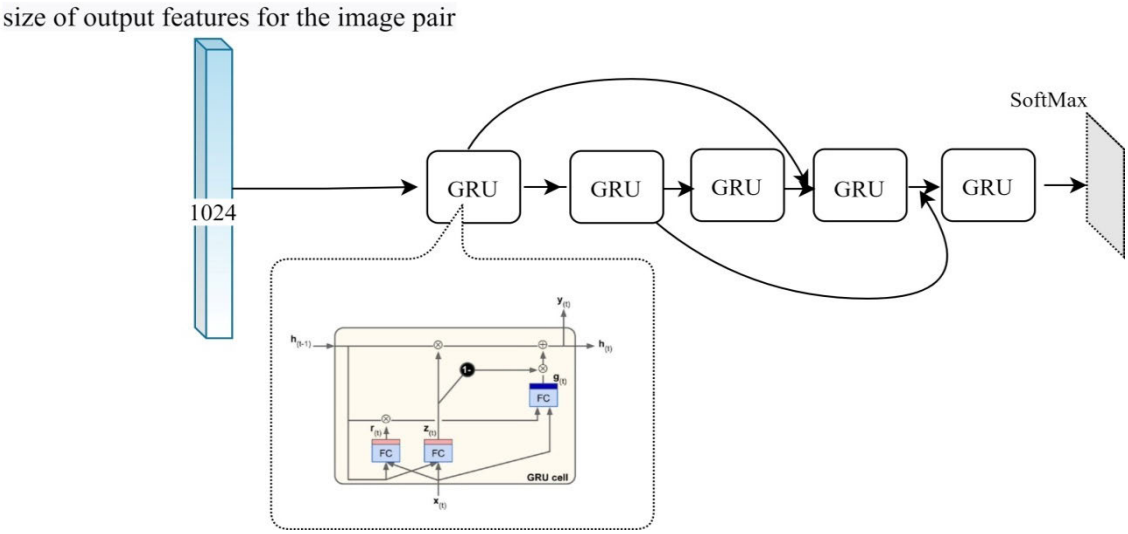


FIGURE 6. Deep-skip-GRU architecture.

deep learning framework with an additional embedding layer to obtain the confidence value, as explained below.

We use a robust correlation filter-based tracker called Minimum Output Sum of Squared Error (MOSSE) tracking filter [36] to record the action sequence of a person or group. This filter can process more than 700 frames per second and is insensitive to changes in lighting, scale, poses, and abrupt movements.

The next step is to extract pyramidal convolutional features from two consecutive images for each tracked person to recognize their activities. Representing the content of visual patterns, especially in videos, is challenging. In this pipeline, we use an end-to-end deep visual flow architecture called LiteFlowNet [35] to reinforce our proposed framework. NetC and NetE are two subnetworks in this architecture. The NetE network consists of cascading flow inference units and modulation units that estimate coarse to fine flow fields. The NetC network, on the other hand, transforms a pair of images or frames from an incoming video chain into two hierarchies of high-dimensional and multi-level information. In other words, while NetE constructs the optical flow between two images using DE convolutional filters, NetC extracts feature descriptors using convolutional filters. After processing the two photos individually to extract the semantic features of each image with NetC, the correlation layer combines the semantic representations of both images to show the difference between them, as shown in Fig. 5.

We use a deep skip connection GRU network with auxiliary connections between GRU units instead of fully auto-connected GRU layers, as shown in Fig. 6. The deeper structure of LSTM or GRU models generally leads to the vanishing gradient problem, which is overcome in this

framework by inserting skip connections between GRU layers. It has been shown that this shortcut connection avoids the vanishing gradient problem during training and leads to significantly better performance [35].

According to the equations of GRU:

$$h_t^j = \left(1 - z_t^j\right) h_{t-1}^j + z_t^j h_t^{\vee j} \tag{7}$$

$$z_t^j = \sigma(W_z x_t + U_z h_t)^j \tag{8}$$

$$h_t^{\vee j} = tanh(W x_t + U(r_t \circ h_{t-1}))^j \tag{9}$$

$$r_t^j = \sigma(W_r x_t + U_r h_t)^j \tag{10}$$

In (7), the h_t^j activation of each jth GRU unit at time step t is calculated using the linear interpolation of h_{t-1}^j, the hidden state at the previous time step, and $h_t^{\vee j}$, the activation of the current hidden state. In our case, $h_t^j = 1024$ features are extracted from two consecutive video frames at time t from the LiteFlowNet architecture. The parameter z_t^j of (8) is the update gate and works as a special kernel that decides how much the activation of each unit needs to be updated. The parameter $h_t^{\vee j}$ in (9) is the hidden state at the current time step computed as in traditional RNNs. Moreover, in (10) the parameter r_t^j is the reset gate, and when it is close to 0, it effectively forgets the information of the previously computed state.

The pipeline predicts object movements in image series to differentiate between people running or walking. Labels from the output are used as keywords in training, enhancing existing knowledge for accurate decision-making (Fig. 7).

C. FEEDBACK LOOP AND FUSION OF PIPELINES OUTPUTS

The weighted average plays a crucial role in determining the overall score of abnormality in the arrangement of frames. This technique is used in the fusion stage to concatenate the weights obtained from the first pipeline (representing the likelihood of anomalies) with the weights obtained from the second pipeline (representing the confidence of predicted captions). The first pipeline detects changes in frames by assigning scores based on movement, with higher scores for more significant movements. The second pipeline extracts captions from frames, assigns scores to these captions, and activates a trigger if the first pipeline score exceeds a predefined threshold. An initial weight is assigned to the scores from both pipelines to determine their relative importance in the weighted average calculation. The weighted average technique, applied through (11), calculates a new weight, representing the overall abnormality level of the frame (see Fig. 2 and Fig. 8). This new weight is expressed as a linguistic variable for use in the next step, and is also fed back to the first pipeline to initiate an iterative loop or feedback loop that repeats the process, potentially influencing the subsequent results. The weighted averaging process is performed iteratively for each frame, considering scores from both pipelines and updating the weights accordingly. Each weight calculation produces a linguistic variable, representing the abnormality level based on information from both pipelines. When a frame with a low score is encountered, indicating lower abnormality, the trigger for the second pipeline to end is activated, as further analysis may not be necessary. Ultimately, all weight rates obtained through the weighted average technique are computed to derive a comprehensive linguistic variable for each video clip, capturing the overall abnormality level of the clip.

$$W = \sum_{i=1}^{n} w_i X_i / \sum_{i=1}^{n} w_i \tag{11}$$

In the (11), W represents the weighted average, n represents the number of scores being averaged, w_i represents the weights applied to the scores, and X_i represents the scores themselves.

In our case, the weighted average W will be:

$$W = (w_1 X_1 + w_2 X_2)/(w_1 + w_2) \tag{12}$$

where w_1 and w_2 are the weights assigned to the anomaly scores and captions, and X_1 and X_2 represent the respective scores and captions.

For example (suppose the frame number is 1):

Frame 1: Score from the first pipeline = 0.7, Weight = 0.6.

Frame 1: Score from the second pipeline = 0.6, Weight = 0.4.

Weighted Average: $(0.7 \times 0.6) + (0.6 \times 0.4) / (0.6 + 0.4) = 0.6$. The new weight obtained from the weighted average is 0.6, represented as a linguistic variable indicating a moderate level of abnormality.

Feedback Loop: This new weight (weighted average) is sent back to the first pipeline for further processing, potentially influencing subsequent scores and weights.

To demonstrate the advantages of weighted average integration, let us consider a scenario where the first pipeline detects an anomaly event with a high score in a specific frame. By focusing on detecting and tracking objects associated with this event, the second pipeline produces more precise and pertinent captions. The weighted average integration guarantees that anomaly detection considers both anomaly scores and object-level information from the captions, leading to enhanced performance.

In summary, this fusion of outputs can provide a more comprehensive understanding of the video frames by incorporating anomaly event detection and object-level information.

D. FUZZY SYSTEM PROCEDURES

Most methods for determining if something unusual has occurred in a video are based on binary logic (yes/no or 1/0). In this logic, a decision is made that something unexpected has happened without providing further details. Using fuzzy logic, which allows for more than two answers, can offer more detailed information. These details are crucial in reducing false alarms in the output. In fuzzy logic, fuzzy linguistic variables must be defined to represent abnormality scores, along with a fuzzy function that relies on these variables. This paper utilizes the weighted average to create linguistic

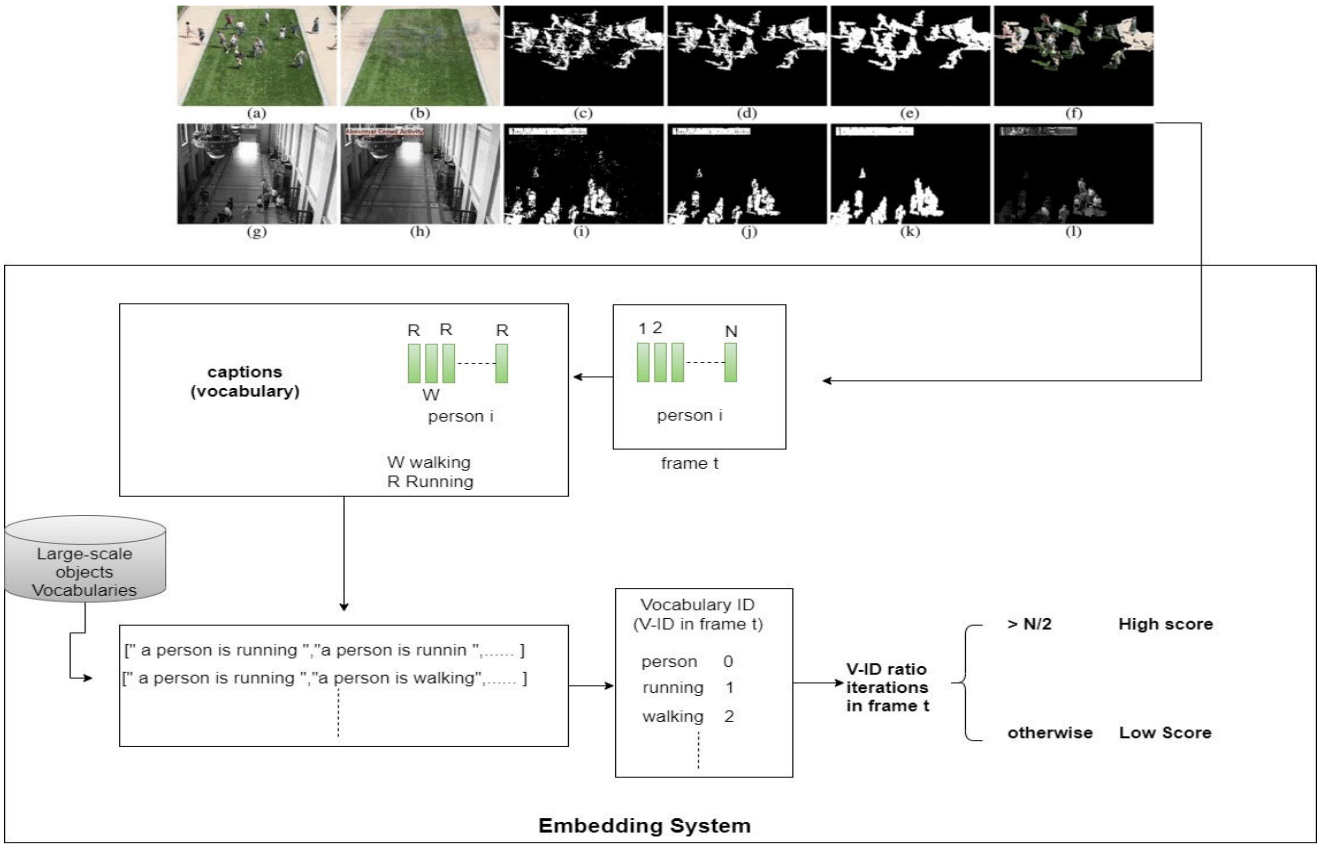


FIGURE 7. Overview of caption-based classification by embedding system.

variables that signify scores of abnormality. Linguistic variables offer a more intuitive and understandable representation of abnormal events. For instance, low, medium, and high linguistic variables can be established based on specific ranges of the weighted average.

Define the linguistic variable for the first pipeline outcome as ''abnormal'' with corresponding membership function scores as follows:

- Low Anomaly (0.3-0.5).
- Medium Anomaly (0.6-0.7).
- High Anomaly (0.7-1).

The procedure for estimating the truth value of this variable is as follows:

Input S: Set of segments, each containing a set of K frames, and within each j frame, there are n position points at time point t_j, denoted as $(x_1^j, y_1^j), \ldots, (x_n^j, y_n^j)\}$ set of ROIs, where n represents the number of objects.

Step 1: As discussed in Section 3.1, the K frames are divided into two epochs (normal and abnormal) based on their confidence score using a DSCNN-based neural network developed from scratch.

Step 2: Determine the truth value of the target area by mapping the confidence score to the interval [0, 1].

The *normal* linguistic variable is similar to the anomaly linguistic variable, except the threshold is set less than 0.3.

The linguistic variable for the second pipeline outcome is either ''running'' or ''walking'' with the following membership function scores:

- Running (0.3-1).
- Walking (0-0.2).

Input S: Same anomaly linguistic variable segments, but (only negative segments or when the trigger is on).

Step 1: At time pointt_j, the average value between all possible point pairs is determined by detecting and tracking a set of $(x_1^j, y_1^j), \ldots, (x_n^j, y_n^j)\}$. As mentioned in Section 3.2, if the average value is greater than $n/2$, the set of these pairs receives the running label based on their confidence score.

Step 2: Estimate the true value of the running by mapping the confidence score to the interval [0, 1].

The walking variable is described as the running variable, but in our case, the average value between all possible point pairs will be less than 0.5.

Finally, the Anomaly-epoch fuzzy function is defined at time t_j as follows:

$$\begin{aligned}&\text{Anomaly}(t_j)\\&= [[\text{anomaly}(O_{object(i)}, t_j) \vee \text{normal}(O_{object(i)}, t_j)]\\&\quad \wedge [\text{running}(O_{object(i)}, t_j) \vee \text{walking}(O_{object(i)}, t_j)]].\end{aligned} \tag{13}$$

where O_{object} is the position of each tracked person.

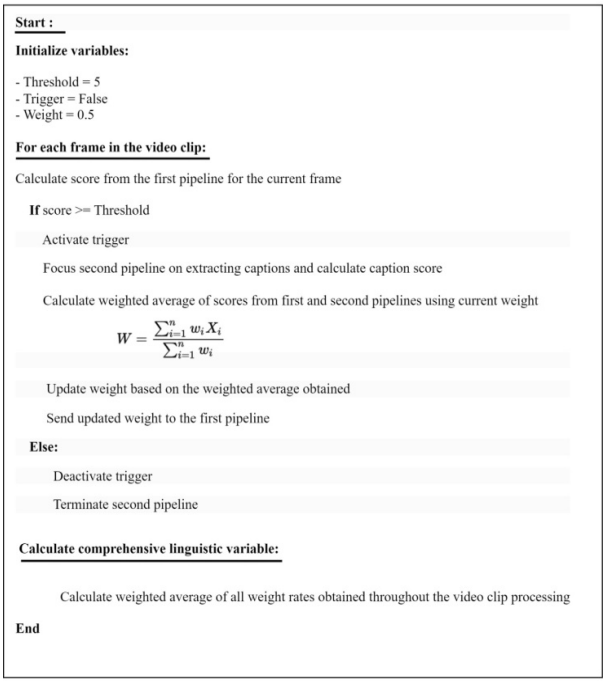


FIGURE 8. Scores fusion process.

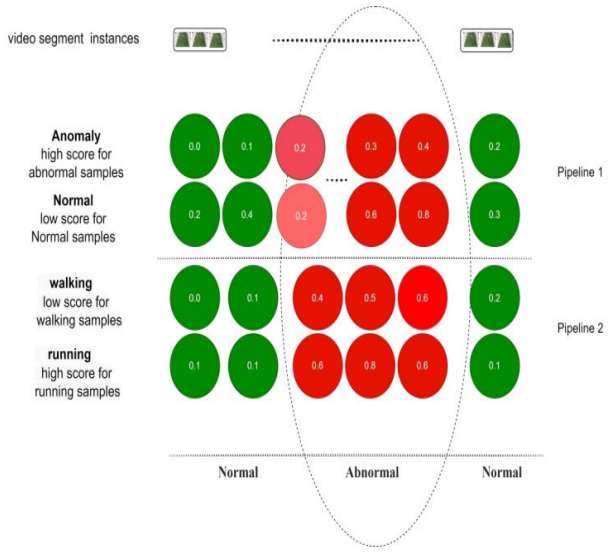


FIGURE 9. Descriptive illustration of the proposed method outcomes.

These steps are depicted in the Fig. 9 as a descriptive illustration. The red region in this figure represents the region where abnormal conditions are activated.

V. EXPERIMENTS AND ANALYSIS

A. IMPLEMENTATION DETAILS

We evaluated our method on well-known crowd abnormality datasets in a Google Colab notebook using Keras, sci-kit-learn, and Python libraries. We also used the following hardware configuration for these experiments: an Intel Core i7-4960x processor running at 3.6 GHz, 8 GB of Kingston DDR3-1600 MHz RAM memory, a Seagate 1.5 TB hard drive, and a Geforce Titan X graphics card. We compared our findings to the state of the art and conducted the evaluation using a frame-level protocol in a standard configuration.

B. DATASETS

For the preliminary experimental phase of our proposed system, we used the crowd video dataset UMN [37]. This dataset includes a group of people initially walking normally in different directions, then suddenly moving quickly and behaving abnormally in different directions, as if they are running away from something. In this case, we have two different time intervals are separated by classification and a predetermined threshold. Due to the limited availability of previous datasets, we collected our dataset by searching YouTube and iStock [38] videos and using textual search queries (such as "panic," "accident escape," and "police and arrests") for each type of anomaly. We also used text queries in various languages (Arabic, English, etc.) to retrieve as many videos as possible for each anomaly. We used these datasets to test our method. We gathered untethered videos from various real-world locations, including escapes from explosions or assaults, stress and sudden escapes due to robberies and police arrests, casual videos of groups of people in public places, and some videos we filmed ourselves. Many of these videos have been added to the UMN dataset in Fig. 10 due to their impact on public safety in our community. For training, we need video-level labels to configure the dataset. We need to know the temporal annotations (the start and end frames of the anomaly) in each anomaly video to evaluate performance in the test videos. Our dataset is divided into two parts: a training set of 200 normal video segments with 1000 to 3000 frames each, and 175 abnormal video segments with panic or running or other high probability of abnormality, with 1000 to 2000 frames each, and a test set of 50 normal videos and 40 abnormal videos.

During the training phase, a segment of frames from the UMN dataset (see Fig. 10 (a)) is simultaneously delivered to both pipelines. The initial pipeline analyzes the frame and utilizes the DWS-CNN network to assign a score to categorize it into the correct class. As the components of the initial pipeline are functioning, the components of the second pipeline are analyzing the same frame (entered in the first pipeline) with the LiteFlowNet detector and a DSC-GRU network to recognize the objects (in our case, the persons) and evaluate their actions. In other words, the two pipelines collaborate on the same UMN input segment. Our goal was to maintain uniformity and enhance understanding of the characteristics and elements of our framework. To assess how well our model performed, we gathered an additional dataset distinct from the one used for training, as shown in Fig. 10 (b). This dataset contains videos depicting various scenarios related to unusual crowd conduct. The videos were gathered from numerous sources and websites to guarantee a wide range of content and relevance.

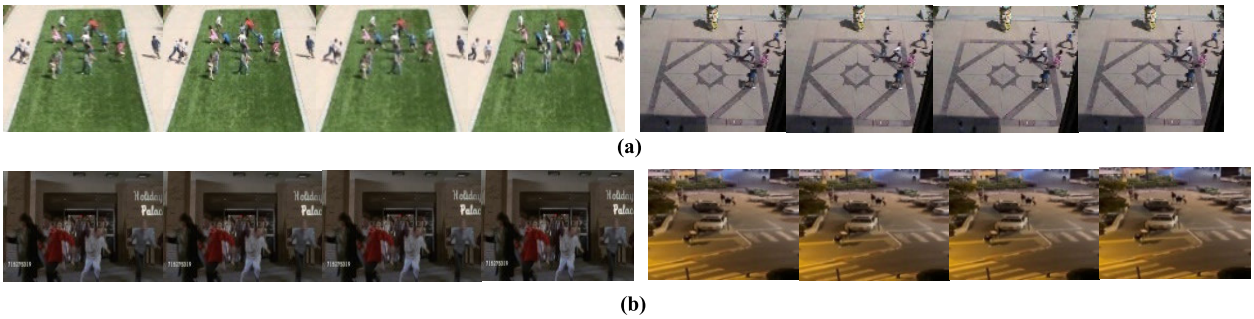


FIGURE 10. Examples of different anomaly samples used in our proposed method.

```
Algorithm. 1 create an optimizer instance
    x-train
    y-train
    training-epoch = 20
    Batch-Size = 64
    Validation_split=0.2
    Dataset = DS
        Step 1: loop over the batches in  DS
                For  x-train ,y-train  in DS
                # forward pass
                   y^ = model(x)
                #  loss for this batch
                   Loss_value
                # obtaining gradients
                    gradients ←·   gradient  (Loss_value , trainable_weights)
                update the weights
                step 1 to finish training-epoch
```

FIGURE 11. Optimizer process.

```
Algorithm. 2  learning process with variable value for learning-rate
    x-train
    y-train
    training-epoch = 20
    Batch-Size = 64
    Validation_split=0.2
    Dataset = DS

        #object of optimizer class
           Optimizer ←— adam (learning rateof 0.001 and Beta_1 =0.9 , beta_2 = 0.999, clipnorm=0.1)
        #compile our model
           Our_Model ←— ( loss function , optimizer , metrics )
        #make a changeable Learning-rate during learning process.
        #factor patience min_lr
           Lr_scheduler ←— (factor = 0.0001 , patience=2 , min_lr = 0.00001 )
        #balancing classes weights
           Classes_weights ←— number of classes ( balanced )
        # training our model
           Our_model ←— fit :  x_train, y_train, Batch_size, training-epochs
           Validation_split=0.2
```

FIGURE 12. The learning process.

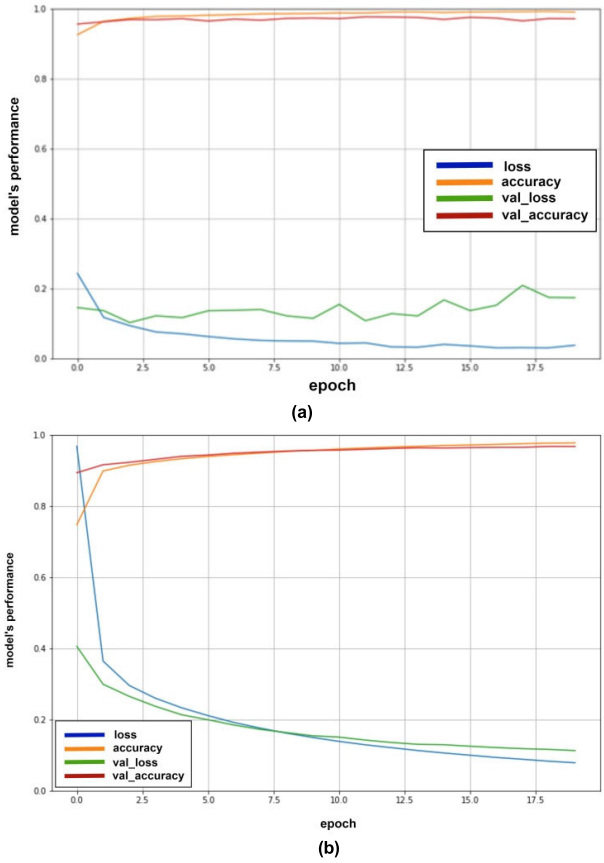


FIGURE 13. The first pipeline results on various datasets. a) On the UMN dataset with a fix learning rate b) On the UMN dataset with variable learning rate.

C. EXPERIMENTAL SETUP AND IMPLEMENTATION DETAILS

*The Pre-Processing Module:*Images are read from the dataset using the Glob library to retrieve image links, processed through OpenCV to standardize their sizes to (224, 224), normalized to [0,1] range, converted to floats with numpy, and split into training/testing sets using Train_Test_Split.

*Experimental Results Regarding the Detection Accuracy of Abnormal Segments (the First Pipeline):*During the training process, we divide the training set into small batches and feed them into the proposed network in pipeline 1. The network consists of two parts: one for feature extraction and another for receiving a flattened vector of features to perform the classification process. To reduce the multiplications, we replace these layers with depthwise separable convolution (which performs 25 times fewer operations than the original convolution [39]). We collect the features for the frames in batches, followed by max pooling layers. After extensive testing, we select the layers and factors that help reduce overfitting and underfitting. The final features set (4096) is fed into a flattened layer and a fully connected 3-layer network. We use 0.30% dropout regularization and the normal kernel initializer, which ensures that the weights are not

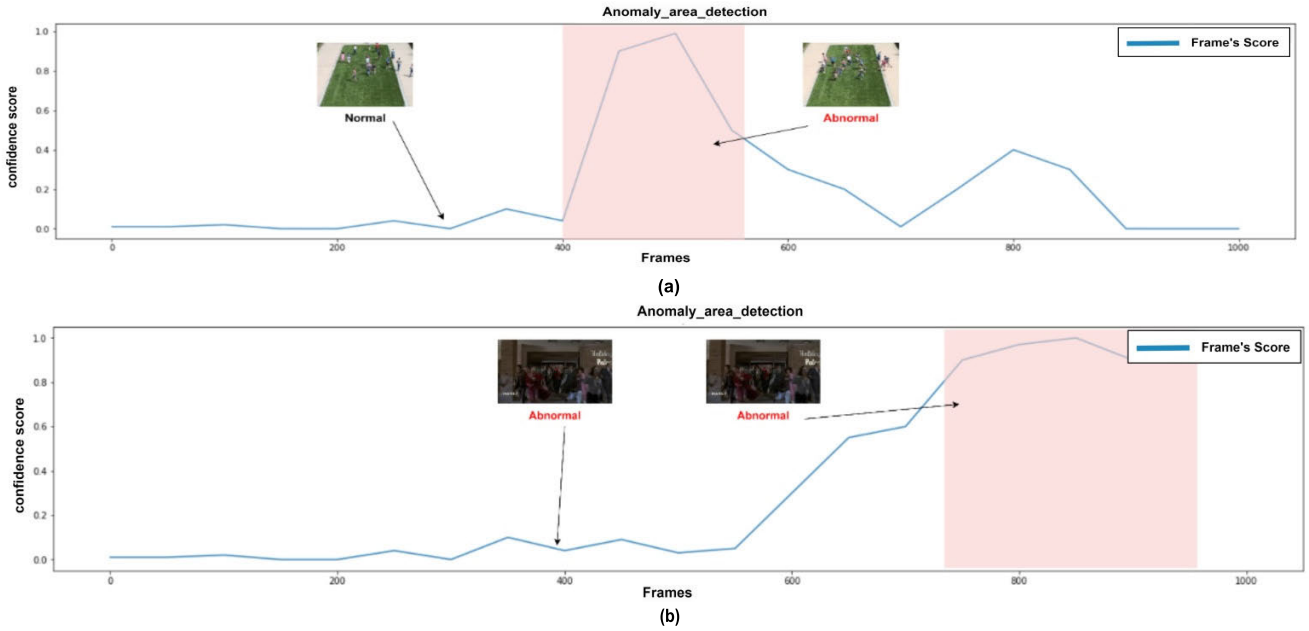


FIGURE 14. Qualitative results on a) the UMN dataset and b) our dataset collection.

TABLE 1. A Specific target classification performance.

Dataset	OBJECT	Precision	Recall
Pedestrian	Specific-Person	34 %	26 %
Highway	Specific-Car	49 %	53 %
Recorded-videos	Specific Person	38 %	32 %

TABLE 2. Target type classification performance.

Dataset	OBJECT	Precision	Recall
Pedestrian	People	56 %	47 %
Highway	Vehicle	65 %	71 %
Recorded-videos	People	58 %	54 %

TABLE 3. An example of the fuzzy attributes (knowledge set) in our method and their related results.

Pipeline-1 (P_1)	PIPELINE-2 (P_2)	Y
Normal	Running persons	Weak Normality
Normal	Walking persons	Normal Activity
Abnormal	Running persons	Strong Abnormality
Abnormal	Walking persons	Weak Normality

random at the start to avoid gradient problems (explosion or vanishing) and to ensure that the network does not migrate to the largest learning set. We use the ReLU activation function and sigmoid activation for the first and last fully connected (FC) layers, respectively, as well as the Adam optimizer (to balance speed and accuracy) with an initial learning rate of 0.001 and Beta 1 = 0.9 and Beta 2 = 0.999 parameters. In the training process, the number of training epochs (the number of times all the data passes through the network; one pass includes a forward and backward pass, and one epoch is one pass) was set to 20 epochs. The training set is divided into two segments. Each segment passes through the network and produces an output. The loss function is then calculated for each element. After calculating the average cost, the kernel initializer sets the initial weights (we used he_norm). The gradient (hyperplane) is then calculated based on the cost and weights. The optimizer then takes the gradient and searches for the appropriate weights and the optimal point. It repeats this process with the next batch from the training epoch set until the learning process is complete and the optimal point is reached, as shown in Fig. 11.

We have placed restrictions on the gradient and learning rate that the optimizer uses to move on the surface of the gradient to reach the best point and avoid the problem of gradient explosion. Because the features are normalized (to values of 0 or 1), and the gradient should not exceed 1, so we set a specific threshold (gradient clipping) using the ClipNorm property when creating the optimizer in Keras to keep the gradient within a limited range. To avoid skipping the best point (the optimal point), we also set a variable learning rate using a function that adjusts the learning rate at specific numbers of training epochs. The learning process is described in pseudo-code form in Fig. 12.

The network proposed in the first pipeline was tested on the UMN dataset in its three segments (the small playground, inside the store, and in the courtyard), and on a set of videos we created. Fig. 13 shows the results of the proposed network on datasets with variable learning rates. We observe that when the learning rate is variable, the results of classifying abnormal segments are improved.

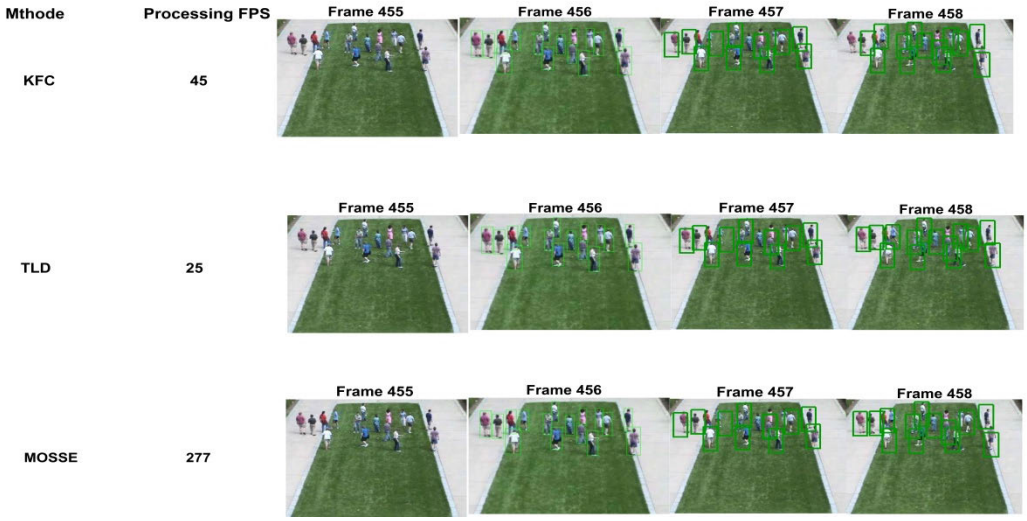


FIGURE 15. Results were achieved on the UMN dataset using various tracking algorithms.

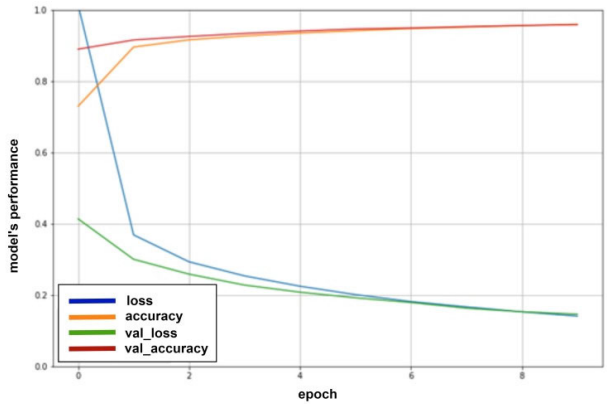


FIGURE 16. DS-GRU Sequential patterns are learned with 10 training epochs.

TABLE 4. Comparison of Scores with and without weight average technique.

Status	Average Score (With Weight Average)	Average Score (Without Weight Average)	Score rate changing	Improvement
Abnormal	0.56	0.54	higher is better	+0.02
Running	0.60	0.57	higher is better	+0.03
Normal	0.32	0.34	lower is better	+0.02
Walking	0.33	0.34	lower is better	+0.01

Fig. 14 (a) reflects some of the qualitative results, with a failure case in Fig. 14 (b) due to a lack of clarity in our dataset videos.

Experimental Results Regarding Detection and Tracking ROI (the Second Pipeline): The same resized frames from our training epochs are fed into Pipeline 2 concurrently. We have selected a pedestrian dataset with 1099 frames with a dimension of 360 × 240, the majority of which are pedestrians, and a highway dataset with 1700 frames with a dimension of 320 × 240, the majority of which are vehicles. 150 images are used as test data, and 400 cropped images from each dataset are used as training data. The experiments' findings for the mentioned datasets and a few of our recorded video segments are presented in Table 1. Table 1 shows the detection precision of a specific target, such as a specific person. For detecting specific targets, such as a moving person, high-resolution images of the target's distinctive features are required. For instance, pedestrians frequently have a height-to-width ratio greater than one, whereas vehicles typically have an aspect ratio smaller than one. The detection results in Table 2, therefore, outperform those in Table 1.

After detecting a pedestrian in the video stream, we use the MOSSE tracker to track the individual throughout the video. We compared various tracking techniques and found that MOSSE offers quick and accurate results for activity recognition. Fig. 15 shows the visual comparisons of different tracking algorithms and their processing rates in frames per second. TLD [40] and KCF [41], as well as other advanced methods, tend to drift in fast motion, while MOSSE has shown better results with a high FPS, making it suitable for real-time activity recognition.

The Deep Spatio-Temporal Gated Recurrent Unit (DS-GRU) is a type of neural network that is trained to extract features from a video clip by analyzing pyramidal convolutional features from two consecutive frames of a tracked individual using the LiteFlowNet CNN. These features, comprising 1024 values from two consecutive frames are fed into the DS-GRU at a specific time step. The DS-GRU has 256 hidden dense units and is trained for 10 epochs with a dropout of 0.3 and an initial learning rate of 0.001. The performance of the DS-GRU on our collected dataset is shown in Fig. 16. The captions generated by the GRU are fed

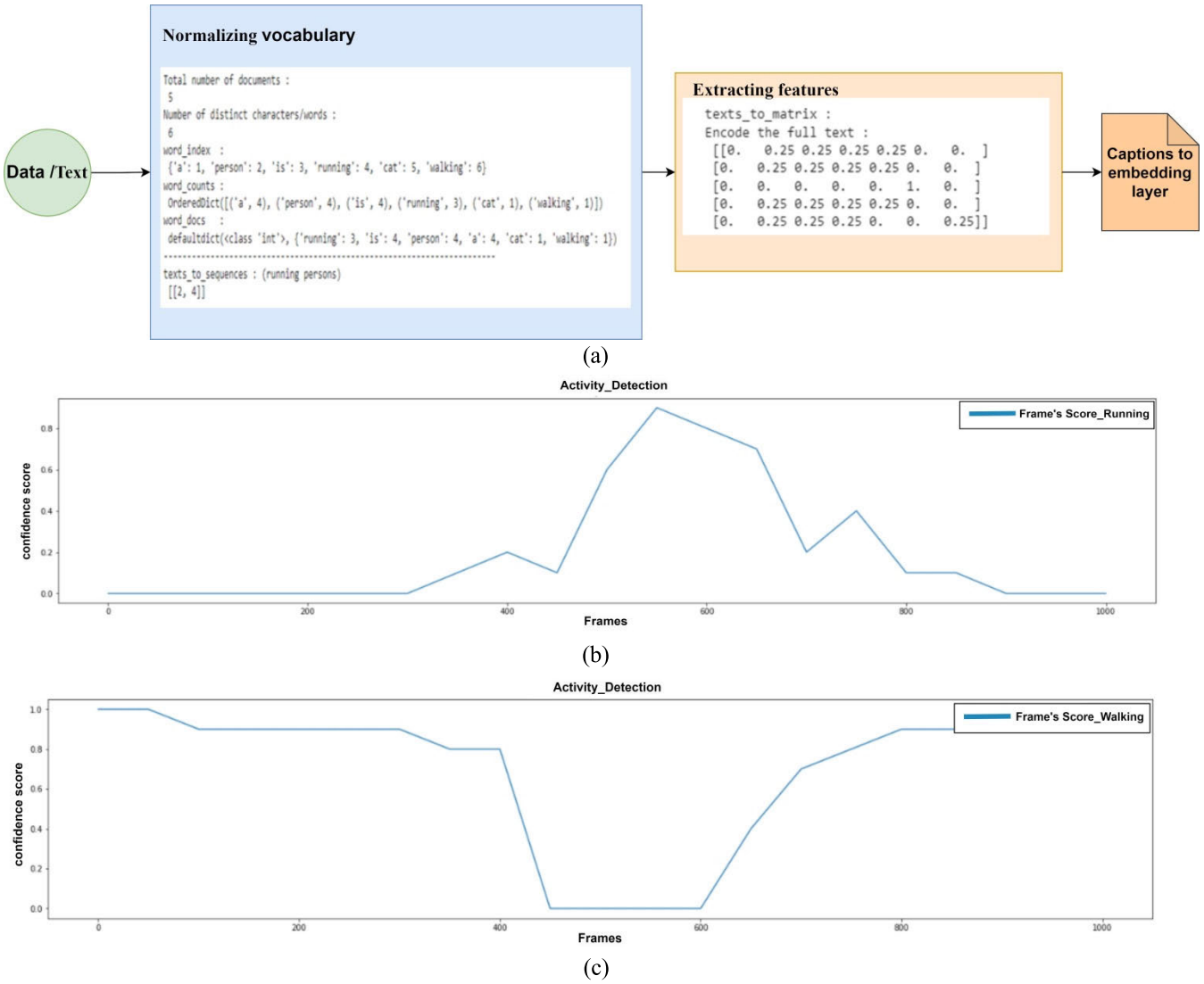


FIGURE 17. The process of preparing text data and its outcome.

into an embedding system, which analyzes them and assigns confidence scores to them.

Preparing the Text Data: To prepare the text data, we extracted captions for each frame and split the words using the *tokenizer* Class. It enabled us to build a vocabulary and transform the text into sequences with unique identifiers for each element. In Fig. 17(a), the extracted features from this text sample are shown along with the frequency of each item. We normalized the vocabulary during the training phase by converting it to numbers and accounting for the frequency of each word (for instance, "a person runs," "someone runs," "a person runs," and "a person walks"). By doing so, we can recognize the most prevalent activity in a frame, like a person running. After preparing the data, we divided it into training data and labels for each text, where the labels indicate whether the data belongs to the normal or abnormal class. The data was then fed into the embedding layer, followed by a GRU network with a sigmoid function to obtain results. To evaluate our method, we used precision, recall, and F1-score metrics. Fig. 17(b) and (c) show the running and walking areas, respectively, from frame 1 to frame 1000.

After the results from the previous pipelines have been obtained, we can use rules and fuzzy logic assignment functions to determine the correct area of the video in which anomalies occur. By utilizing multiple sources in decision-making, the results are considered that have already been extracted and then a final decision is made using fuzzy logic and its rules. This approach helps us to make more accurate and reliable decisions.

Consider N as the Number of Attributes, K as the Number of Rules, and if the universe of discourse to be

$\mathbf{X} = x1, x2, x3, \ldots.,$, so the *kth* fuzzy rule is:

$$R_k : if\ (x_1 = F_{k1} \wedge x_2 = F_{k2} \wedge \ldots \wedge x_N = F_{kN})\ then (y = B_k)$$

where $B_k, F_{kj}, k = 1, 2, 3, \ldots, K; j = 1, 2, 3, \ldots, N$ are linguistic values.

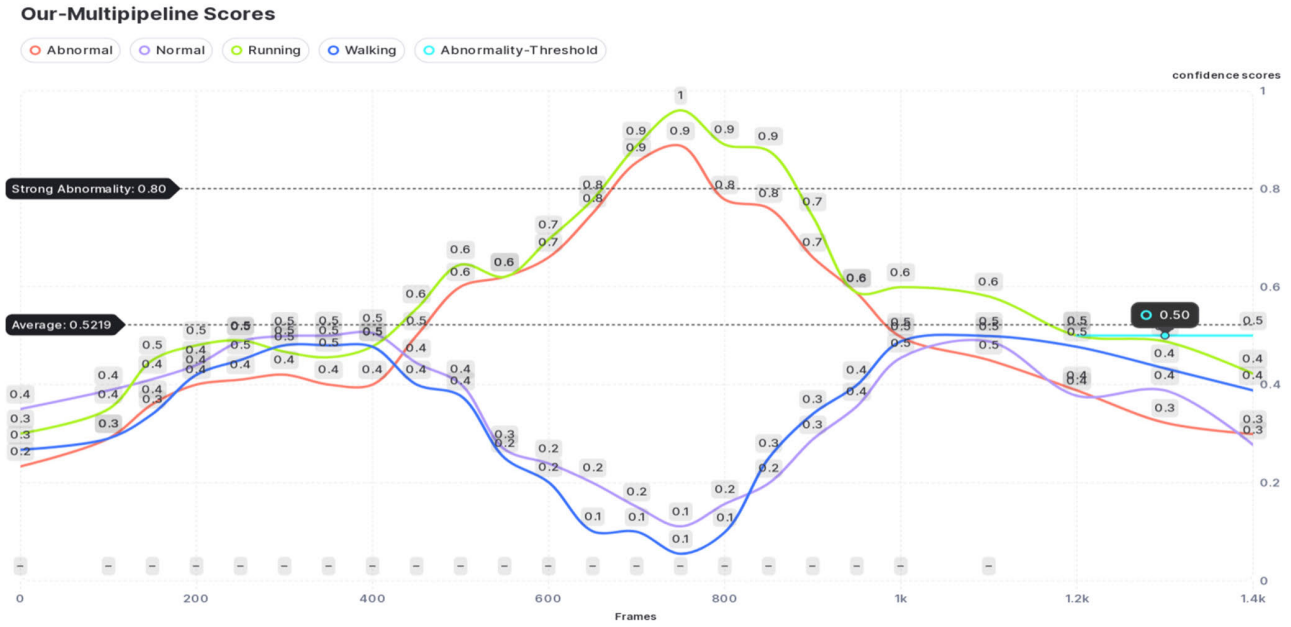


FIGURE 18. Results obtained from our proposed method without weighted average.

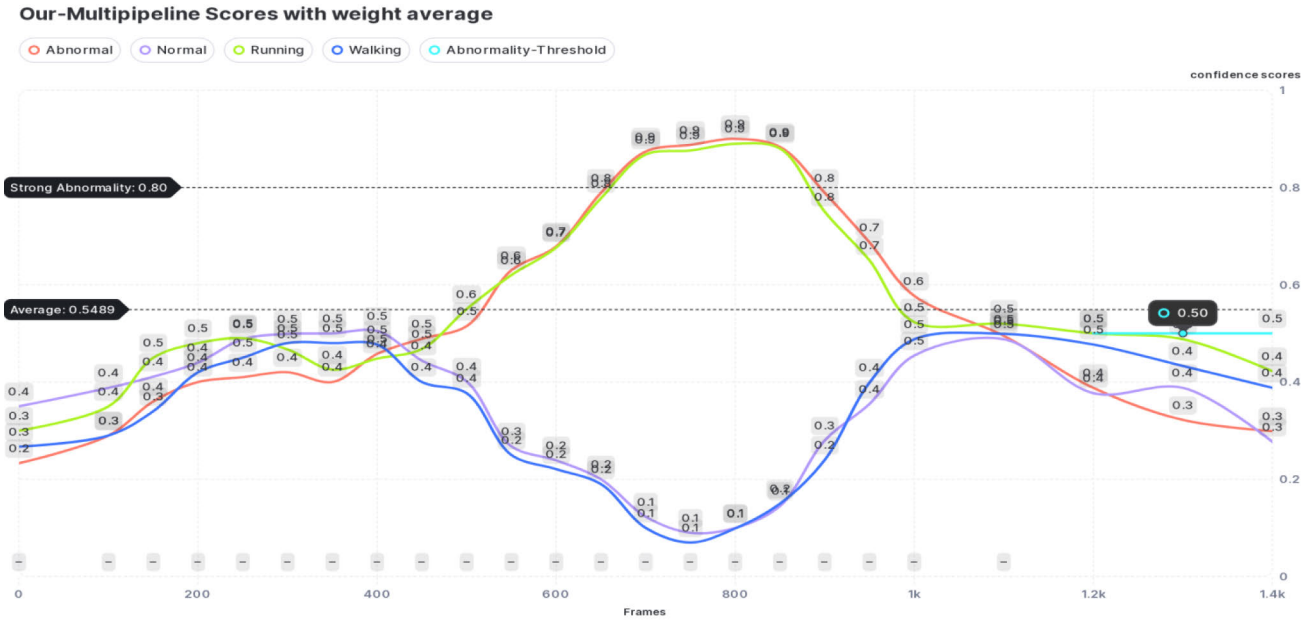


FIGURE 19. The results obtained from our proposed method with weighted average.

In our case (due to paper constraints, we will only consider one scenario): As stated in Section 3.4:

N: Abnormal, Normal, Running, Walking.

Output (y)=Anomaly-epoch (t_j)=[Abnormal ($O_{object(i)}$, t_j) Normal($O_{object(i)}$, t_j)] $\wedge$ [Running ($O_{object(i)}$, t_j) Walking ($O_{object(i)}$, t_j)]].

The accuracy of fuzzy logic in producing reliable results depends on the appropriate selection of predicates or attributes. For example, if the results of the predicates from the three pipelines (the knowledge set) are as follows: *anomaly* area (abnormal epoch) and *running* persons, then our final logical answer would be "*Strong Abnormality*".

Additional examples of some of the outcomes of the proposed fuzzy logic-based method are provided in Table 3.

D. DATA ANALYSIS AND RESULTS

The data analysis performed on the UMN dataset focused on improving the detection of anomalies in surveillance, with

TABLE 5. Comparison of performance of proposed approach with existing techniques on UMN dataset.

method	Actual start frame(Scene 1/2/3)	Start frame (Scene 1/2/3)	AUC Scene 1/2/3	AUC total
annotated UMN dataset video	481 / 1799 / 6700	588 / 1900 / 6998	-	0.955
Proposed in [42]	481 / 1799 / 6700	610 / 1988 / 6985	-	0.887
Proposed in [25]	481 / 1799 / 6700	580 / 1888 / 6870	0.96 / 0.94 / 0.96	0.968
Proposed in [43]	481 / 1799 / 6700	515 / 1850 / 6880	0.97 / 0.96 / 0.97	0.970
Proposed in [16]	481 / 1799 / 6700	500 / 1750 / 6550	0.97 / 0.95 / 0.95	0.968
Proposed in [17]	481 / 1799 / 6700	525 / 1800 / 6900	0.97 / 0.96 / 0.96	0.970
Our proposed method	481 / 1799 / 6700	477 / 1700 / 6600	0.97 / 0.97 / 0.96	**0.972**
Our proposed method with Weight Average	481 / 1799 / 6700	485 / 1820 / 6790	0.98 / 0.98 / 0.97	**0.989**

particular emphasis on the Running and Abnormal cases. The weighted average technique provided valuable insights into the performance of different statuses and contributed to the efficiency of anomaly detection in crowded scenes.

We obtained scores for each video frame, which served as indicators of the events occurring within the frame. Fig. 18 visually presents the results obtained from our proposed method, showcasing the scores predicted from the frames using the two pipelines explained earlier. By assigning specific scores to each frame to express the presence of abnormal situations and considering other scores to represent actual activity within the same frame, we definitively confirm the occurrence of abnormal events. Crossing the abnormality threshold triggered the detection of abnormal events through fuzzy processes, and the scores for each linguistic variable allowed for the classification of the abnormality level of these events. However, we observed some instability in the Running case, resulting in incorrect decisions. Despite the lack of a linear relationship between the Abnormal and Running statuses, our model's results proved its effectiveness by comparing them with previous research.

A new method utilizing the scores extracted for each frame as linguistic variables in a fuzzy system enabled accurate decision-making based on these variables.

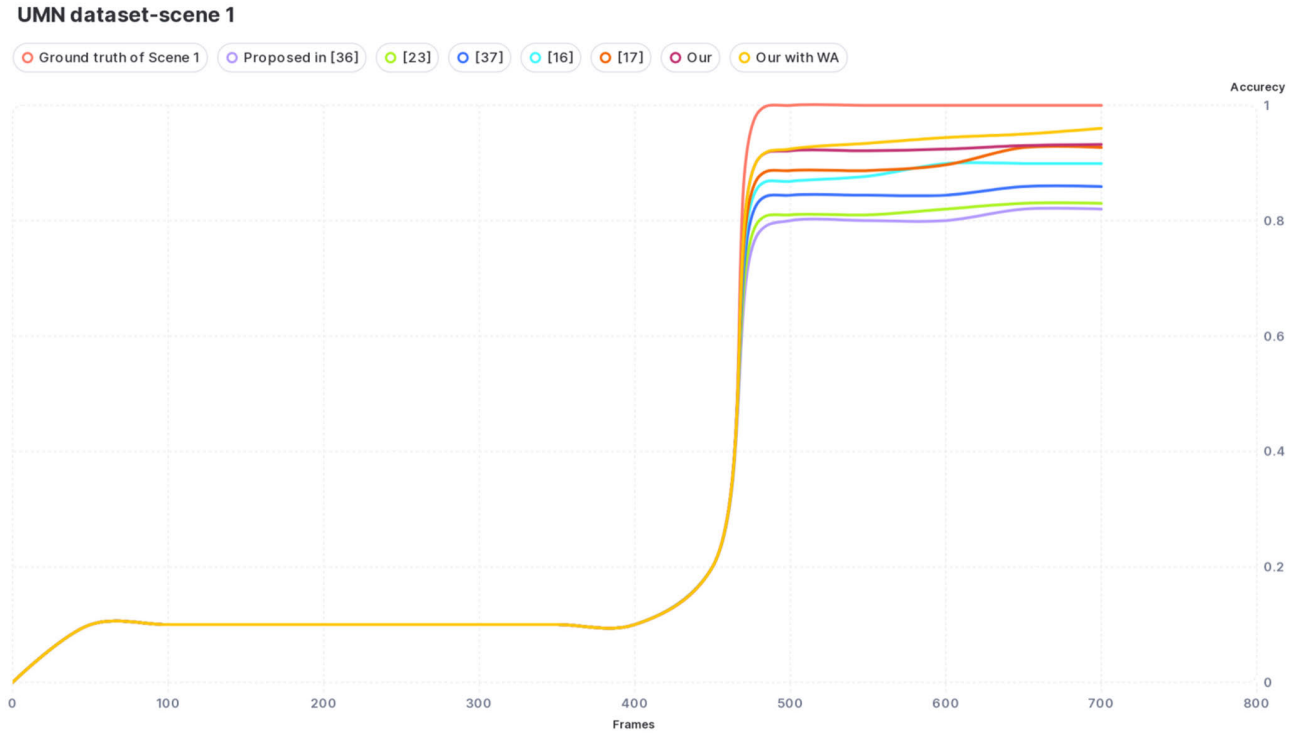
To address the issue of false alarms and differences between the scores of Abnormal and Running, we aimed to establish a relationship between them, ensuring their scores remained close over time. We adjusted the scores to establish a connection between the Abnormal and Running cases using the weighted average technique. As depicted in Fig. 19, when the abnormality score exceeded the threshold, the weighted average technique enhanced consistency and similar scores between the two statuses during the anomaly event. It highlighted a significant level of relative stability in these cases.

The analysis showed that the abnormal case had an average score of 0.5 around frame 490, closely matching the actual onset of anomalies in the first scene of the UMN dataset. The scores then increased, reaching 0.9 by frame 800, indicating the presence of significant anomalies. Likewise, the running case had a high average score of 0.89 at frame 800, slightly lower than the abnormal case. It corresponds to the actual video, where individuals are observed running randomly between frame 650 and frame 850.

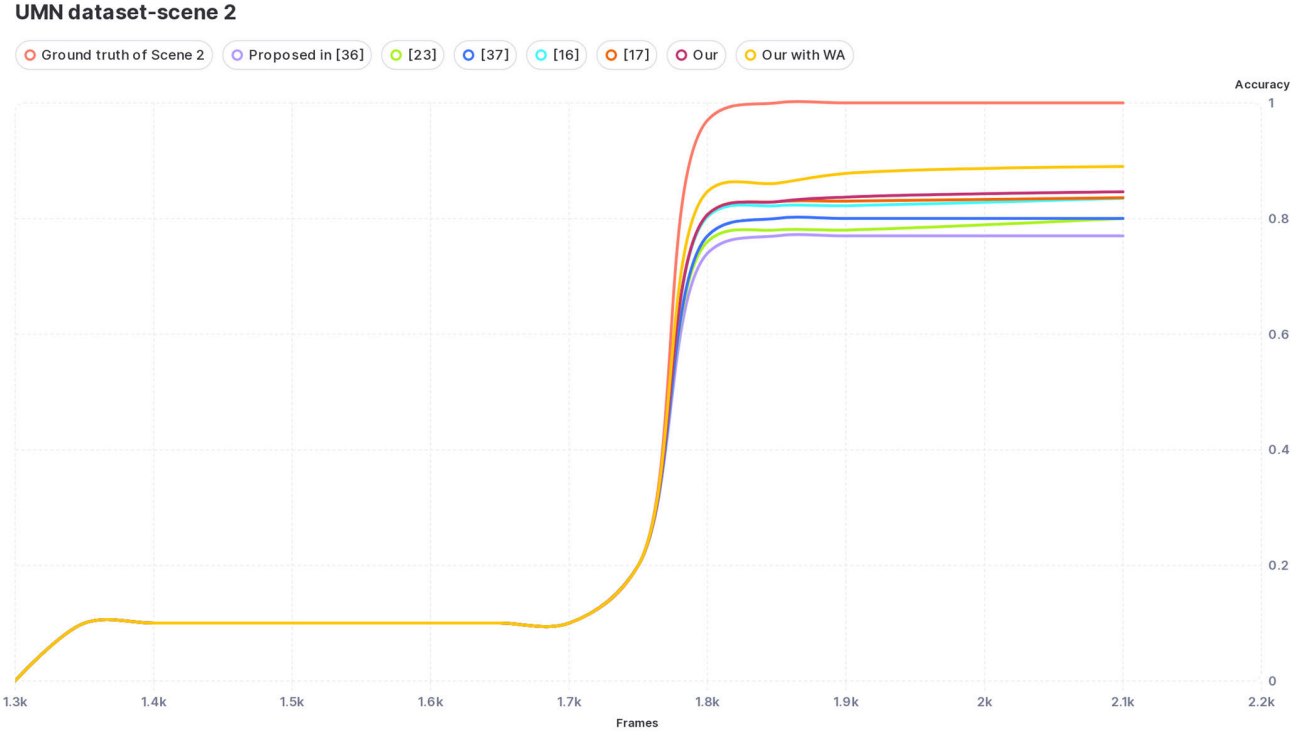
Table 4 compares the average scores obtained with and without the weighted average technique. The scores for Abnormal and Running cases remained relatively similar between the two approaches, indicating the stability and consistency of the results. However, the weighted average technique improved the average scores for both statuses by $+0.01$. It signifies a more accurate representation of the detected anomalies within the frame range of 480 to 900. The scores for the Normal and Walking cases remained consistent in both approaches, indicating their lower occurrence of anomalies.

In conclusion, our data analysis on the UMN dataset underscored the significance of focusing on the Running and Abnormal cases for improving anomaly detection in crowded scenes. The utilization of the weighted average technique enhanced the accuracy and reliability of the results, providing valuable insights into the performance of different statuses and contributing to the overall efficiency of surveillance anomaly detection systems.

Performance of Proposed Approach on UMN Dataset: The results in Figures 20 and 21 indicate that our proposed method, with and without the Weighted Average (WA) technique, outperforms the other evaluated methods in anomaly detection for surveillance. The inclusion of the WA technique noticeably enhances the performance of our method. Specifically, at the 490th frame mark, our method with WA achieves



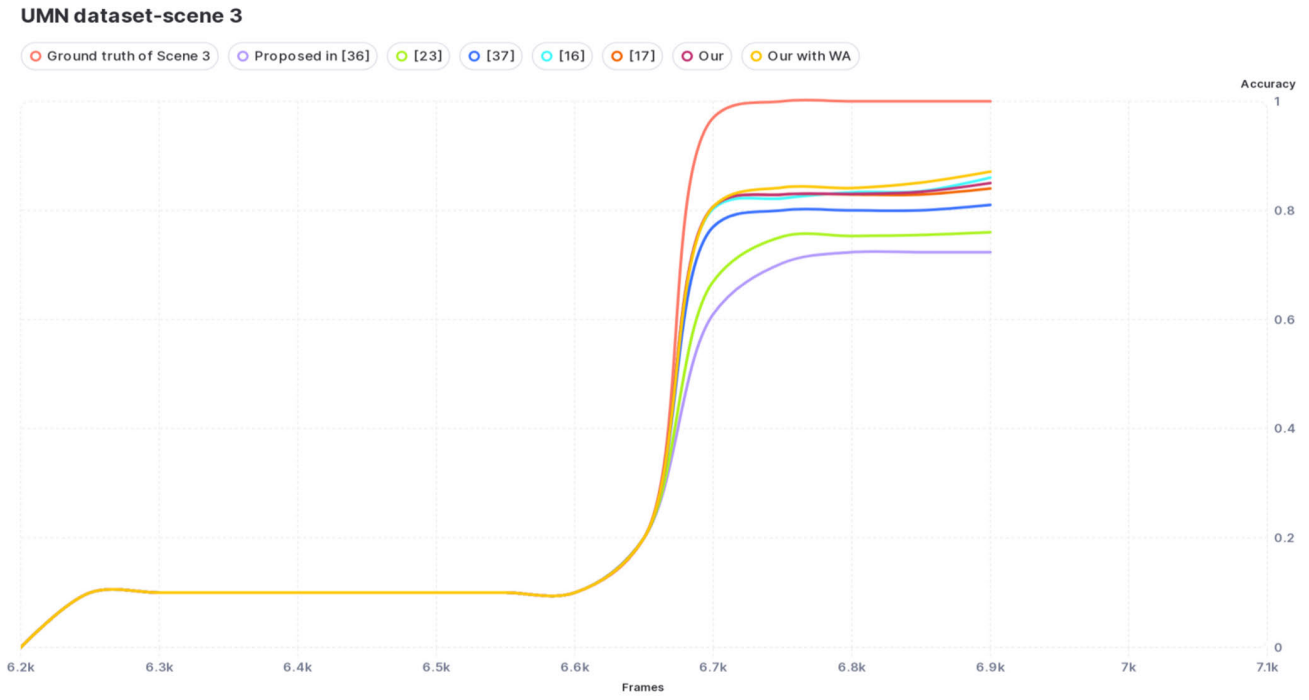
(a)



(b)

FIGURE 20. Performance of the proposed approach compared with other techniques on the UMN dataset. (a) Performance on Scene 1 of the dataset. (b) Performance on Scene 2 of the dataset.

a 0.003 higher detection rate compared to the method without WA, demonstrating the effectiveness of the WA technique in achieving more accurate anomaly detection. Furthermore, our proposed method consistently achieves higher scores



(c)

FIGURE 20. *(Continued.)* **Performance of the proposed approach compared with other techniques on the UMN dataset. (c) Performance on Scene 3 of the dataset.**



FIGURE 21. **Performance of the proposed approach compared with other techniques on all of the UMN dataset scenes.**

than the cited methods, particularly in the later stages of the analysis. Noteworthy performance peaks are observed at frame marks 480, 1800, and 6750, with scores reaching 0.98, 0.987, and 0.972, respectively, as shown in Table 5. These highlight the superior anomaly detection capabilities of our method.

VI. CONCLUSION

Sharing information can help increase the number of sources providing diverse perspectives, leading to more accurate and comprehensive answers. In line with this principle, our research introduces a high-level information-sharing approach for detecting movement patterns and identifying abnormal crowd behavior in surveillance videos. This approach leverages fuzzy predicates to represent information and concepts, employing specific fuzzy logic functions based on knowledge, action-reaction interpretation, and changes in patterns within actual video sequences. To further enhance the accuracy and reliability of our approach, we incorporated the Weighted Average (WA) technique. By calculating weighted averages of detection scores, the WA technique effectively captures the collective wisdom from multiple frames, improving the overall anomaly detection performance. The WA technique significantly enhances the detection rate, as demonstrated by the 0.003 increase at the 490 frame mark compared to the method without WA. These results highlight the effectiveness of the WA technique in achieving more precise anomaly detection. The fusion of fuzzy predicates inferred from the two pipelines, and their associated confidence scores forms the foundation of our fuzzy logic functions. The assignment functions for these predicates are determined based on the interpretation of the video sequence content. Through evaluation using ground truth annotations obtained from actual video sequences, our proposed approach has shown promising and encouraging results.

In conclusion, our high-level method, improved with the Weighted Average technique, offers significant benefits for detecting movement patterns and identifying abnormal crowd behavior in surveillance videos. The integrated WA technique results in improved detection rates, enhancing the accuracy and reliability of our approach. The successful evaluation of our proposed method using actual video sequences and ground truth annotations further underscores its potential for practical implementation in real-world surveillance scenarios.

Despite successes, the proposed method faces challenges and limitations, primarily relying on the precision of detectors and trackers, which, if inaccurate, can highly impact overall accuracy. Accurate annotations for training and evaluation are necessary, but hard to obtain in real-world surveillance scenarios. Additionally, the method's current execution demands high computational resources for its processing stages, potentially limiting real-time use in resource-constrained environments. In addition, the method's scope is currently limited to detecting running and walking patterns, with the potential for expansion to cover a broad range of anomaly behaviors like loitering and other types of interactions, enhancing its applicability.

In future work, we will focus on enhancing the detector in the second pipeline by improving its accuracy and implementation speed. We also aim to expand the proposed approach to encompass the detection of all actions that occur (our current work was limited to running and walking) and to identify the environment by analyzing its background and introducing its attributes in the second pipeline. Additionally, we plan to extend the approach to include sound analysis for diagnosing abnormal events in the scene (such as the sound of an explosion followed by a running condition, used to confirm an abnormal condition). Finally, we intend to conduct a complete evaluation of the method on multiple video sequences of actual crowd scenes in various environments.

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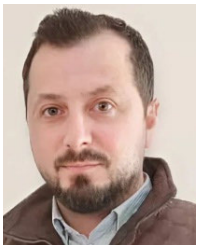
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